

Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB

Objective 1 Matlab Code and Output:

```
clc; clear; close all;

% Simulated soil moisture data (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold for irrigation

threshold = 40;

figure

% ----- Graph 1: Soil Moisture Variation -----

subplot(2,1,1)

plot(time, soil_moisture, '-o', 'LineWidth', 2)

title('Soil Moisture Monitoring in IoT Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold, 'r--', 'Threshold')

% ----- Graph 2: Moisture Status (Dry/Wet Level) -----

status = soil_moisture >= threshold;

subplot(2,1,2)

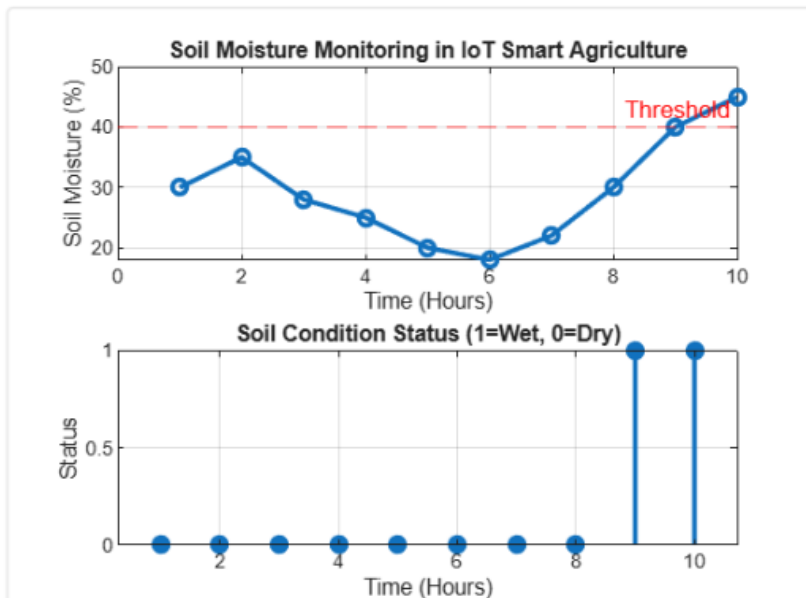
stem(time, status, 'filled', 'LineWidth', 2)

title('Soil Condition Status (1=Wet, 0=Dry)')

xlabel('Time (Hours)')

ylabel('Status')

grid on
```



Objective 2 Matlab Code and Output:clc; clear; close all;

clc;

clear;

close all;

% Simulated soil moisture (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level

threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)

pump_status = soil_moisture < threshold;

figure;

% Graph 1: Soil Moisture vs Threshold

```
subplot(2,1,1);  
plot(time, soil_moisture, '-o', 'LineWidth', 2);  
hold on;  
yline(threshold, 'r--', 'Threshold');  
grid on;  
title('Soil Moisture vs Pump Control');  
xlabel('Time (Hours)');  
ylabel('Soil Moisture (%)');
```

% Graph 2: Pump Status

```
subplot(2,1,2);  
stem(time, pump_status, 'filled', 'LineWidth', 2);  
grid on;  
title('Water Pump Status (1 = ON, 0 = OFF)');  
xlabel('Time (Hours)');  
ylabel('Pump Status');  
ylim([-0.2 1.2]);
```

